Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following pair of orbitals possess two nodal planes

(A) $p_{xy'} d_{x^2-y^2}$

(B) $\mathrm{d_{xy}}, \mathrm{d_{zx}}$

(C) $\mathrm{p_{xy}}, \mathrm{d_{zx}}$

(D) $d_{zz}, d_{x^2-y^2}$

Q.2

Atoms consist of protons, neutrons and electrons. If the mass of neutrons and electrons were made half and two times respectively to their actual masses, then the atomic mass of ${}_6\mathrm{C}^{12}$

(A) Will remain approximately the same

(B) Will become approximately two times

(C) Will remain approximately half

(D) Will be reduced by 25%

Q.3

The correct order of increasing energy of atomic orbitals is

(A) $5p < 4f < 6s < 5d$

(B) $5p < 6s < 4f < 5d$

(C) $\mathrm{4f < 5p < 5\ d < 6\ s}$

(D) $5p < 5d < 4f < 6s$

Q.4

Bohr's atomic model can explain

(A) the spectrum of hydrogen atom only

(B) the spectrum of an atom or io containing one electron only

(C) the spectrum of hydrogen molecule

(D) the solar spectrum

@IITJEE_Advanced

Q.5

Which of the following postulates does not belong to Bohr's model of the atom?

(A) Angular momentum is an integral multiple of $\frac{\mathrm{h}}{2\pi}$

(B) The electron stationed in the orbit is stable

(C) The path of an electron is circular

(D) The change in the energy levels of electron is continuous

Q.6

If the following matter waves travel with equal velocity, the longest wavelength is that of a/an.

(A) electron

(B) proton

(C) neutron

(D) $\alpha-$ particle

Q.7

Which set of quantum numbers is not consistent with the quantum mechanical theory?

(A) $\mathrm{n}=2, l=1,\ \mathrm{m}=1,\ \mathrm{s}=1/2$

(B) $\mathrm{n}=4, l=3,\ \mathrm{m}=2,\ \mathrm{s}=-1/2$

(C) $\mathrm{n}=3, l=2,\ \mathrm{m}=3,\ \mathrm{s}=+1/2$

(D) $\mathrm{n}=4, l=3,\ \mathrm{m}=3,\ \mathrm{s}=+1/2$

Q.8

The quantum numbers of most energetic electron in Ne atom when it is in first excited state is

(A) $2,1,0,\pm 1/2$

(B) $3,1,1,\pm 1/2$

(C) $3,0,0,\pm 1/2$

(D) $3,1,0,\pm 1/2$

Q.9

Of the following transitions in hydrogen atom, the one which gives an absorption line of lowest frequency is

(A) $\mathrm{n}=1$ to $\mathrm{n}=2$

(B) $\mathrm{n}=3$ to $\mathrm{n}=8$

@IITJEE_Advanced

(C) $\mathrm{n}=2$ to $\mathrm{n}=1$

(D) $\mathrm{n}=8$ to $\mathrm{n}=3$

Q.10

If numerical value of mass and velocity are equal then de-Broglie wavelength in terms of KE is

(A) $\frac{\mathrm{mh}}{2\mathrm{KE}}$

(B) $\frac{vh}{2\mathrm{KE}}$

(C) both are correct

(D) none is correct

Q.11

The triad of nuclei that is isotonic is

(A) ${}_{6}^{14}\mathrm{C},{}_{7}^{15}\mathrm{N},{}_{9}^{17}\mathrm{F}$

(B) ${}_{6}^{12}\mathrm{C},{}_{7}^{14}\mathrm{N},{}_{9}^{19}\mathrm{F}$

(C) ${}_{6}^{14}\mathrm{C},{}_{7}^{14}\mathrm{N},{}_{9}^{17}\mathrm{F}$

(D) ${}_{6}^{14}\mathrm{C},{}_{7}^{14}\mathrm{N},{}_{9}^{19}\mathrm{F}$

Q.12

If uncertainty in position and velocity are equal then uncertainty in momentum will be

(A) $\frac{1}{2}\sqrt{\frac{mh}{\pi}}$

(B) $\frac{1}{2}\sqrt{\frac{\mathrm{h}}{\pi\mathrm{m}}}$

(C) $\frac{\mathrm{H}}{4\pi\mathrm{m}}$

(D) $\frac{\mathrm{mh}}{4\pi}$

Q.13

Which of the following electronic transitions requires that the greatest quantity of energy be absorbed by a hydrogen atom ?

(A) $\mathrm{n}=1$ to $\mathrm{n}=2$

(B) $\mathrm{n}=2$ to $\mathrm{n}=4$

(C) $\mathrm{n}=3$ to $\mathrm{n}=6$

(D) $\mathrm{n}=1$ to $\mathrm{n}=\infty$

@IITJEE_Advanced

Q.14

How fast is an electron moving if it has a wavelength equal to the distance it travels in one second?

(A) $\sqrt{\frac{\mathrm{h}}{\mathrm{m}}}$

(B) $\sqrt{\frac{m}{h}}$

(C) $\sqrt{\frac{h}{p}}$

(D) $\sqrt{\frac{\mathrm{h}}{2(\mathrm{KE})}}$

Q.15

If the shortest wavelength of H atom in Lyman series is x, the longest wavelength in Balmer series of $\mathrm{He}^{+}$is

(A) $\frac{9x}{5}$

(B) $\frac{36x}{5}$

(C) $\frac{x}{4}$

(D) $\frac{5x}{9}$

Q.16

When a certain metal was irradiated with light of frequency $3.2 \times 10^{16}\ \mathrm{Hz}$, photoelectrons emitted had twice the kinetic energy as did photoelectrons emitted when the same metal was irradiated with light of frequency $2.0 \times 10^{16}\ \mathrm{Hz}$. Hence, threshold frequency is

(A) $0.8 \times 10^{15}\ \mathrm{Hz}$

(B) $8.0 \times 10^{15}\ \mathrm{Hz}$

(C) $0.8 \times 10^{14}\ \mathrm{Hz}$

(D) $6.4 \times 10^{16}\ \mathrm{Hz}$

Q.17

For a 'p'-electron, the orbital angular momentum is

(A) $\sqrt{6}\hbar$

(B) $\sqrt{2}\hbar$

(C) $\hbar$

(D) $2\hbar$

Q.18

Charge to mass ratio of cathode rays depends on

@IITJEE_Advanced

(A) Nature of the gas in the discharge tube

(B) Pressure of the gas in the discharge tube

(C) Material of cathode

(D) None of these

Q.19

Which of the following radiations has the highest wave number?

(A) X-rays

(B) Microwaves

(C) I.R. rays

(D) Radio Waves

Q.20

Which of the following represents schrodinger equations

(A) $$\left(\frac{\partial^2}{\partial \mathrm{x}^2}+\frac{\partial^2}{\partial \mathrm{y}^2}+\frac{\partial^2}{\partial \mathrm{z}^2}\right) \Psi+\frac{8 \pi^2 \mathrm{~m}}{\mathrm{~h}^2}[\mathrm{E}-\mathrm{V}] \Psi=0$$

(B) $\hat{H}\Psi = E\Psi$

(C) $(\hat{T}+\hat{V})\Psi = E\Psi$

(D) All of them

Q.21

O_2 undergoes photochemical dissociation into 1 normal oxygen atom (O) and more energetic oxygen atom O^*. If $\left(\mathrm{O}^*\right)$ has $1.967 \mathrm{eV}$ more energy (O) and normal dissociation energy of O_2 is $498 \mathrm{~kJ} \mathrm{~mol}^{-1}$, what is the maximum wavelength effective for the photochemical dissociation of O_2 ?

Q.22

The electron energy in hydrogen atom is given by $\mathrm{E}=\left(-21.7 \times 10^{-12}\right) / \mathrm{n}^2$ ergs. By calculating the energy required to remove an electron completely from the $\mathrm{n}=2$ orbit, then the longest wavelength of light that can be used to cause this transition is $\mathrm{p} \times 10^{-6} \mathrm{~cm}$, then the value of p is(round off to the nearest integer)?

Q.23

One mole of He^{+}ion is excited. Spectral analysis showed the existence of 50% ions in $3^{3^{\mathrm{d}}}$ level, 25% in

@IITJEE_Advanced

2nd level and remaining 25% in ground state. Ionization energy of He^{+} is $54.4eV$; the total energy evolved when all the ions return to ground state is $k \times 10^5$, then the value of k is(round off to the nearest integer)

Q.24

The λ of H_α line of the Balmer series is $6500\mathring{A}$. What is the λ of H_β line of the Balmer series ?(round off to the nearest integer)

Q.25

A near UV photon of $300\ nm$ is absorbed by a gas and then re-emitted as two photons. One photon is red with a wavelength of 760 nm. Hence, wavelength of the second photon is (round off to the nearest integer)

Q.26

Wave number of a spectral line for a given transition is $3\ cm^{-1}$ for He^{+}, then its value for Be^{3+} (isoelectronic of He^{+}) for same transition is

Q.27

A standing wave in a string $35\ cm$ long has a total of six nodes, (including those at the ends). Hence, wavelength of the standing wave is

Q.28

To move the electron in one H atom from the ground state to the second excited state, $12.084eV$ are needed. How much energy(in kJ) is needed to cause $1\ mol$ of H atoms to undergo this transition?

Q.29

If the radius of first Bohr orbit is $\frac{8}{\pi}$, then de-Broglie wavelength of electron in 3rd orbit is nearly

Q.30

An atom has 35 nucleons and has an atomic number equal to 17. The number of electrons with $n = 2,\ m = 0$ in it is

Answers & Solutions

Q.1 Answer:

d_{xy}, d_{zx}

Solution:

@IITJEE_Advanced

d_{xy} **and** d_{zx} **has two nodal planes.**
whence $\lambda = 3.67 \times 10^{-5}$ cm

Q.2 Answer:

Will be reduced by 25%

Solution:

No change by doubling the mass of electrons however by reducing the mass of the neutron to half, total atomic mass becomes $6 + 3$ **instead of** $6 + 6$**. Thus reduced by** 25%**.**

Q.3 Answer:

$5p < 6s < 4f < 5d$

Solution:

According to Aufbau principle electron are filled in various atomic orbitals in the increasing order of energy: $1\text{ s} < 2\text{ s} < 2\text{p} < 3\text{ s} < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5\text{ d} < 6\text{p} < 7\text{ s}$

Q.4 Answer:

the spectrum of an atom or io containing one electron only

Solution:

In Bohr's theory while calculating the energy of electrons, the potential energy has been found out by considering only the attraction between the electron and nucleus. If there is another electron in the orbit, the potential energy would change due to electron-electron repulsion. Therefore the Bohr's model is meant for all one-electron systems.

Q.5 Answer:

The change in the energy levels of electron is continuous

Solution:

In Bohr's model, the energy is discrete and not continuous

Q.6 Answer:

electron

Solution:

$$\lambda = \frac{\text{h}}{\text{mv}}$$

λ **will be large if** m **is small**

@IITJEE_Advanced

Q.7 Answer:

$\mathrm{n}=3, l=2, \mathrm{m}=3, \mathrm{s}=+1/2$

Solution:

The e^{-} is a well defined correlation between $\mathrm{n}, l, \mathrm{m}$ and s quantum numbers in an atom. $l=(\mathrm{n}-1), \mathrm{m}$ will be $\pm 1/2$, for $\mathrm{n}=3$ there cannot be $\mathrm{m}=3$. Highest value m will be 2 .

Q.8 Answer:

$3, 0, 0, \pm 1/2$

Solution:

The electronic configuration of Ne atom is $1\ \mathrm{s}^2 2\ \mathrm{s}^2 2\mathrm{p}^6$. In its first excitation state the electron from $2\mathrm{p}$ will unpair and shift to $3\ \mathrm{s}$ where its quantum numbers will be $3, 0, 0, \pm 1/2$

Q.9 Answer:

$\mathrm{n}=3$ to $\mathrm{n}=8$

Solution:

Absorption line in the spectra arise when energy is absorbed i.e., electron shifts from lower to higher orbit, out of a and b, b will have the lowest frequency as this falls in the Paschen series

Q.10 Answer:

both are correct

Solution:

Given $\mathrm{m}=\mathrm{v}$

$$\therefore \quad \lambda=\frac{\mathrm{h}}{\mathrm{mv}}=\frac{\mathrm{h}}{\mathrm{v}^2}$$

Also $\mathrm{KE}=\frac{1}{2}\mathrm{mv}^2$

$$\therefore \quad \mathrm{v}^2=\frac{2\mathrm{KE}}{\mathrm{m}}$$

$$\therefore \quad \lambda=\frac{\mathrm{h}}{\left(\frac{2\mathrm{KE}}{\mathrm{m}}\right)}=\frac{\mathrm{mh}}{2\mathrm{KE}}$$

Q.11 Answer:

$^{14}_{6}\mathrm{C}, ^{15}_{7}\mathrm{N}, ^{17}_{9}\mathrm{F}$

Solution:

@IITJEE_Advanced

All contain 8 neutrons (species containing the same number of neutrons are called isotonic).

Isotonic species are defined as the species which have the same number of neutrons.

No. of neutrons in:

$_6C^{14} = 14 - 6 = 8$

$_7N^{14} = 14 - 7 = 7$

$_9F^{19} = 19 - 9 = 10$

$_7N^{15} = 15 - 7 = 8$

$_9F^{17} = 15 - 7 = 8$

$_6C^{12} = 12 - 6 = 6$

$_6C^{14}, _7N^{15}, _9F^{17}$ **are isotonic species.**

Q.12 Answer:

$\frac{1}{2}\sqrt{\frac{mh}{\pi}}$

Solution:

$\Delta x = \Delta v$

$\Delta x \Delta v = \frac{h}{4\pi m}$

$\Delta v^2 = \frac{h}{4\pi m}$

$\therefore \quad \Delta v = \sqrt{\frac{h}{4\pi m}}$

$\therefore \quad \Delta p = m\Delta v = m\sqrt{\frac{h}{4\pi m}}$

$= \frac{1}{2}\sqrt{\frac{hm}{\pi}}$

Q.13 Answer:

$n = 1$ to $n = \infty$

Solution:

$$\Delta E = \text{ constant } \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

Transition with maximum value of n_2 and least value of n_1 will absorb greatest quantity of energy

Q.14 Answer:

$\sqrt{\frac{h}{m}}$

Solution:

$\lambda = \frac{h}{mv}$ **but** $v = \lambda$

$\therefore \quad \lambda v = v^2 = \frac{h}{m}$

$\therefore \quad v = \sqrt{\frac{h}{m}}$

@IITJEE_Advanced

Q.15 Answer:

$\frac{9x}{5}$

Solution:

$$\frac{1}{\lambda} = \overline{\mathrm{R}}_{\mathrm{H}} \mathrm{Z}^2 \left[\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right]$$

For Lyman series $\mathrm{n}_1 = 1$

λ **is shortest if** $\mathrm{n}_2 = \infty$

$\therefore \overline{\mathrm{R}}_{\mathrm{H}} = \frac{1}{\mathrm{x}}$ **for H-atom** $(\mathrm{Z} = 1)$

For Balmer series, $\mathrm{n}_1 = 2$

λ **is longest if** $\mathrm{n}_2 = 3$

Here $\frac{1}{\lambda_{\max}} = \left(\frac{1}{\mathrm{x}}\right)(2)^2 \left[\frac{1}{2^2} - \frac{1}{3^2}\right]_{\text{for } \mathrm{He}^+ \text{ion}}$

$(Z = 2)$

$\therefore \lambda_{\max} = \frac{9x}{5}$

Q.16 Answer:

8.0×10^{15} Hz

Solution:

$\mathrm{hv} = \mathrm{hv}_0 + (\mathrm{KE})$

$\mathrm{KE} = \mathrm{h}\left(\mathrm{v} - \mathrm{v}_0\right)$

$(\mathrm{KE})_1 = \mathrm{h}\left(3.2 \times 10^{16} - \mathrm{v}_0\right)$

$(\mathrm{KE})_2 = \mathrm{h}\left(2.0 \times 10^{16} - \mathrm{v}_0\right)$

Given $(\mathrm{KE})_1 = 2(\ \mathrm{K.E.})_2$

$\mathrm{H}\left(3.2 \times 10^{16} - \mathrm{v}_0\right) = 2\ \mathrm{h}\left(2.0 \times 10^{16} - \mathrm{v}_0\right)$

$\therefore \mathrm{v}_0 = 8.0 \times 10^{15}$ Hz

Q.17 Answer:

$\sqrt{2}\hbar$

Solution:

Orbital angular momentum

$$= \sqrt{l(l+1)}\hbar$$

$= \sqrt{1(1+1)}\hbar = \sqrt{2}\hbar$

Q.18 Answer:

None of these

Solution:

@IITJEE_Advanced

Charge to mass ratio of cathode rays is independent of the nature of the gas or material of the cathode.

The cathode rays have constant charge to mass ratio (e/m). This is because the nature of cathode rays does not depend upon the gas filled in the discharge tube, rather cathode rays are emitted by the cathode in the tube.

Q.19 Answer:

X-rays

Solution:

Electromagnetic radiation is classified into types according to the frequency of the wave. These types include, in order of increasing frequency, radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays.

The frequency is the inverse of wavelength. Hence, in the order of increasing wavelength, the waves are Gamma rays $(<1\ \mathrm{nm})$**, X-rays** $(1-10\ \mathrm{nm})$**, infrared rays** $\left(700-10^{5}\ \mathrm{nm}\right)$**, microwaves** $\left(10^{5}-10^{8}\ \mathrm{nm}\right)$ **radio waves** $\left(>10^{8}\ \mathrm{nm}\right)$

As the wavenumber is directly proportional to the frequency and is inversely proportional to the wave length, the more the wave length, the less the wave number

Q.20 Answer:

All of them

Solution:

Schrodinger wave equation is:

$$\left(\frac{\partial^{2}}{\partial \mathrm{x}^{2}}+\frac{\partial^{2}}{\partial \mathrm{y}^{2}}+\frac{\partial^{2}}{\partial \mathrm{z}^{2}}\right)\Psi+\frac{8\pi^{2}\,\mathrm{m}}{\mathrm{h}^{2}}[\mathrm{E}-\mathrm{V}]\Psi=0$$

$$\hat{\mathrm{H}}\Psi=\mathrm{E}\Psi$$

$$(\hat{\mathrm{T}}+\hat{\mathrm{V}})\Psi=\mathrm{E}\Psi$$

The Schrà¸£à¸–dinger equation is a partial differential equation that describes the dynamics of quantum mechanical systems via the wave function. The trajectory, the positioning, and the energy of these systems can be retrieved by solving the Schrà¸£à¸–dinger equation.

Q.21 Answer:

—

Solution:

$$\mathrm{O}_{2}\rightarrow \mathrm{O}+\mathrm{O}^{*}$$

Dissociation energy $=498\ \mathrm{kJ\ mol}^{-1}$

$=\frac{498\times1000}{6.02\times10^{23}}$ **J molecule** $^{-1}$

$=8.27\times10^{-19}\ \mathrm{Jmolecule}^{-1}$

Excitation energy to form O^{*}

@IITJEE_Advanced

$= 1.967 \mathrm{eV atom}^{-1}$

$= 1.967 \times 1.6 \times 10^{-19}$ J atom $^{-1}$

$= 3.15 \times 10^{-19}$ J atom $^{-1}$

Total energy $= 11.42 \times 10^{-19}$ J

$\mathrm{E} = \frac{\mathrm{hc}}{\lambda}$

$\lambda = \frac{\mathrm{hc}}{\mathrm{E}}$

$= \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{11.42 \times 10^{-19}}$

$= 174 \times 10^{-9}$ m

$= 174$ nm

Q.22 Answer:

—

Solution:

$\Delta \mathrm{E} = \mathrm{E}_{\infty} - \mathrm{E}_2 = [0] - \left[-\frac{21.7 \times 10^{-12}}{4}\right]$

Since

$\mathrm{E_n} = -\frac{21.7 \times 10^{-12}}{\mathrm{n}^2} \mathrm{ergs}, = \frac{21.7 \times 10^{-12}}{4}$

$\Delta \mathrm{E} = \mathrm{hv}$

$\Delta \mathrm{E} = \frac{\mathrm{hc}}{\lambda}$

$\frac{21.7 \times 10^{-12}}{4} = 6.627 \times 10^{-27} \times \frac{3 \times 10^{10}}{\lambda}$

Q.23 Answer:

—

Solution:

$(\Delta \mathrm{E})_{3 \rightarrow 1} = (54.4) \frac{\mathrm{N}_0}{2} \left[\frac{1}{1^2} - \frac{1}{3^2}\right]$

for $\frac{\mathrm{N}_0}{2}$ **ions falling to ground state**

$= 54.4 \times \frac{4 \times \mathrm{N}_0}{9} \mathrm{eV}$

$(\Delta \mathrm{E})_{3 \rightarrow 1} = (54.4) \frac{\mathrm{N}_0}{4} \left[\frac{1}{1^1} - \frac{1}{2^2}\right]$

and

for $\frac{\mathrm{N}_0}{4}$ **ions falling to ground state**

$= 54.4 \times \frac{3 \times \mathrm{N}_0}{16} \mathrm{eV}$

Hence, total energy

$= 54.4 \times \mathrm{N}_0 \left[\frac{4}{9} + \frac{3}{16}\right] = 54.4 \times 6.02 \times 10^{23} \times \frac{91}{144} \mathrm{eV}$

$= 54.4 \times 6.02 \times 10^{23} \times \frac{91}{144} \times 1.6 \times 10^{-19}$ J

$= 3.31 \times 10^{6}$ J

Q.24 Answer:

—

Solution:

@IITJEE_Advanced

For H_α lines of Balmer series $n_1 = 2, n_2 = 3$

For H_β line of Balmer series $n_1 = 2, n_2 = 4$

$$\therefore \quad \frac{1}{\lambda_{H_\alpha}} = R_H\left[\frac{1}{2^2} - \frac{1}{3^2}\right] \quad \ldots (1)$$

$$\text{and} \quad \frac{1}{\lambda_{H_\beta}} = R_H\left[\frac{1}{2^2} - \frac{1}{4^2}\right] \quad \ldots (2)$$

By Eqs. (1) and (2)

$\frac{\lambda_\beta}{\lambda_\alpha} = \frac{\frac{1}{4} - \frac{1}{9}}{\frac{1}{4} - \frac{1}{16}}$

$\lambda_\beta = \lambda_\alpha \times \left[\frac{80}{108}\right] = 6500 \times \frac{80}{108}$

$= 4814.8 \mathring{A}$

Q.25 Answer:

—

Solution:

Energy values are additives

$$E = E_1 + E_2$$

$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$

$\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

$\lambda = 300 \text{ nm}, \lambda_1 = 760 \text{ nm}$

$\therefore \lambda_2 = 495.6 \text{ nm}$

$\approx 496 \text{ nm}$

Q.26 Answer:

—

Solution:

$\overline{v}(\text{ wave number }) = \overline{R}_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\overline{v}_1 \left(He^+, Z = 2\right) = \overline{R}_H (2)^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\overline{v}_2 \left(Be^{3+}, Z = 4\right) = \overline{R}_H (4)^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\frac{\overline{v_2}}{\overline{v_1}} = 4$

$\therefore \overline{v_2} = 4\overline{v}_1$

$\overline{v_2} = 4^*3 = 12$

@IITJEE_Advanced

Q.27 Answer:

—

Solution:

For six nodes, $\mathrm{n}+1=6, \mathrm{n}=5$

$$\therefore l=\frac{\mathrm{n}\lambda}{2} ; \lambda=\frac{2 l}{\mathrm{n}}=\frac{2 \times 35}{5}=14 \mathrm{~cm}$$

Q.28 Answer:

—

Solution:

$$\begin{aligned}
\Delta \mathrm{E} & =12.084 \mathrm{eV} \\
& =12.084 \times 1.6 \times 10^{-19} \mathrm{~J} \text{ per atom} \\
& =12.084 \times 1.6 \times 10^{-19} \times 6.02 \times 10^{23} \mathrm{~J} \mathrm{~mol}^{-1} \\
& =1164 \mathrm{~kJ}
\end{aligned}$$

Q.29 Answer:

—

Solution:

$mvr_n = \frac{nh}{2\pi}$
$\mathrm{mv}\left(\frac{\mathrm{n}^2 \mathrm{a}_0}{\mathrm{Z}}\right)=\frac{\mathrm{nh}}{2 \pi}$
$\mathrm{a}_0=\frac{8}{\pi}=$ **Bohr's radius of H-atom** $\mathrm{n}=3$
$\mathrm{Z}=1$
$\mathrm{mv}\left(\frac{9}{1}\right)\left(\frac{8}{\pi}\right)=\frac{3h}{2\pi}$
$mv=\frac{h}{48}$
$\lambda=\frac{h}{mv}=48$

Q.30 Answer:

—

Solution:

Electronic configuration : $1 \mathrm{~s}^2, 2 \mathrm{~s}^2, 2 \mathrm{p}^6, 3 \mathrm{~s}^2, 3p^5$ **.2 electrons in** 2 **s and 2 electrons in 2 p-subshell have** $\mathrm{n}=2, m=0$**.**

@IITJEE_Advanced